FACULTY OF COMPUTER APPLICATIONS

Marwadi University
Marwadi Chandarana Group
NAAC A+

Faculty of Computer Applications

MCA Sem-1
(AY-2026-27)

Lab Book

Sub Code: 05MC0108
(DBMS)

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2	Write a PL/SQL block to accept product name, qty and price from user and then calculate discount in Rs. based on the given (%).
3	Write a PL/SQL block to calculate the total, percentage and grade of student based on his/her Rollno from RESULT table. (Create RESULT table with Rollno, Name, Sub1, Sub2, Sub3, Sub4, Sub5, Total, Per, Grade attributes with appropriate data type).
4	1. Write a program that prints value 1 to 100 numbers using FOR LOOP. 2. Write a program that prints value 1 to 100 number using LOOP Command. 3. Write a program that prints value 1 to 100 number using WHILE LOOP Command
5	Write a PL/SQL block which displays gross salary of employees as per user input EID. (Consider EMP table with EID, EName, Deptno,Deptname Gender, Age, BasicSal) with appropriate data types.) Gross_Salary: BASICSAL + (DA + HRA + Medical) – PF. Rules: HRA = 15% of basic, DA = 50% of basic, Medical = Rs. 500, PF = 10% of basic.
6	Write a PL/SQL block which accepts measurement in feet and displays it in cm, inch and meter.
7	Write a PL/SQL block which converts temperature from Celsius to Fahrenheit.
8	Write a PL/SQL block to delete the record of employee for given EID.
9	Write a PL/SQL block that calculates the simple interest based on given principal amount, rate of interest and number of years.
10	Write a PL/SQL block to calculate the square and cube of the given number.

1. Write a PL/SQL block that find:

1. Area of Circle
2. Area of Rectangle
3. Circumference of Circle

```
/*
	Write a PL/SQL block that find
	1.Area of circle(pi*radius*radius)
	2.Area of Rectangle(length*width)
	3.Circumference of circle(2*PI*RADIUS)
*/

ACCEPT CHOICE PROMPT "Enter Your Choice: 1.Area of circle ,2.Area of Rectangle
,3.Circumference of circle "
SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF

DECLARE
	CHOICE NUMBER:=&CHOICE;
	PI NUMBER(7,2):=3.14;
	RADIUS NUMBER(5);
	LENGTH NUMBER(5);
	WIDTH NUMBER(5);
BEGIN
	--CHOICE:=&CHOICE;

	DBMS_OUTPUT.PUT_LINE('    ');
	IF (CHOICE=1) THEN
		RADIUS:=&RADIUS;
		DBMS_OUTPUT.PUT_LINE('Area of circle is : ' ||PI*(POWER(RADIUS,2)));
	ELSIF (CHOICE=2) THEN
		LENGTH:=&LENGTH;
		WIDTH:=&WIDTH;
		DBMS_OUTPUT.PUT_LINE('Area of Rectangle is : ' ||LENGTH*WIDTH);
	ELSIF (CHOICE=3) THEN
		DBMS_OUTPUT.PUT_LINE('Circumference of circle is : ' ||2*PI*RADIUS);
	ELSE
		DBMS_OUTPUT.PUT_LINE('Invalid Choice');
```

```
    END IF;
END;
/
SET SERVEROUTPUT OFF
SET VERIFY ON
SET FEEDBACK ON
```

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p1.sql
Enter Your Choice: 1.Area of circle ,2.Area of Rectangle ,3.Circumference of circle :-  1
Enter value for radius: 12
Enter value for length: 8
Enter value for width: 5

Area of circle is : 452.16
```

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p1.sql
Enter Your Choice: 1.Area of circle ,2.Area of Rectangle ,3.Circumference of circle :-  2
Enter value for radius: 12
Enter value for length: 45
Enter value for width: 40

Area of Rectangle is : 1800
```

```
SQL>  @ D:\Kunjal\DBMS\Unit_1\p1.sql

Enter Your Choice: 1.Area of circle ,2.Area of Rectangle ,3.Circumference of circle :-  3
Enter value for radius: 10
Enter value for length: 40
Enter value for width: 45

Circumference of circle is : 62.8
```

2. Write a PL/SQL block to accept product name, qty and price from user and then calculate discount in Rs. based on the given (%).

```
/*
  Write a PL/SQL  block to accept product name ,qty and price from user
   and then calculate discount in Rs.based on the given (%)
*/

SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF

ACCEPT PN PROMPT "Enter product name :- "
DECLARE
        P_NAME          VARCHAR(20);
        QTY     NUMBER(5);
        PRICE   NUMBER(7,2);
        TP      NUMBER(7,2);
        DISCOUNT        NUMBER(5);
        FP      NUMBER(7,2);
BEGIN
        P_NAME:='&PN';
        QTY:=&QTY;
        PRICE:=&PRICE;
        DISCOUNT:=&DISCOUNT;

        TP:=PRICE*QTY;
        FP:=TP-(TP*DISCOUNT/100);
        DBMS_OUTPUT.PUT_LINE('    ');
        DBMS_OUTPUT.PUT_LINE('Product name is := '||P_NAME);
        DBMS_OUTPUT.PUT_LINE('Total qty is := '||QTY);
        DBMS_OUTPUT.PUT_LINE('Price is := '||PRICE);
        DBMS_OUTPUT.PUT_LINE('Total Price is := '||TP);
        DBMS_OUTPUT.PUT_LINE('Discount is := '||DISCOUNT);
        DBMS_OUTPUT.PUT_LINE('Final Price is := '||FP);

END;
/
SET SERVEROUTPUT OFF
```

SET VERIFY ON
SET FEEDBACK ON

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p2.sql
Enter product name :- key-board
Enter value for qty: 3
Enter value for price: 200
Enter value for discount: 12

Product name is := key-board
Total qty is := 3
Price is := 200
Total Price is := 600
Discount is := 12
Final Price is := 528
```

3. Write a PL/SQL block to calculate the total, percentage and grade of student based on his/her Rollno from RESULT table. (Create RESULT table with Rollno, Name, Sub1, Sub2, Sub3, Sub4, Sub5, Total, Per, Grade attributes with appropriate data type).

```
/*
 Write a PL/SQL block to calculate the total, percentage and grade of
student based on his/her Rollno from RESULT table. (Create RESULT
table with Rollno, Name, Sub1, Sub2, Sub3, Sub4, Sub5, Total, Per,
Grade attributes with appropriate data type).
*/

SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF
DECLARE
	 ROLL_NO NUMBER(5);
	 NAME VARCHAR2(30);
	 SUB1 NUMBER(3);
	 SUB2 NUMBER(3);
	  SUB3 NUMBER(3);
	 SUB4 NUMBER(3);
	 SUB5 NUMBER(3);
	TOTAL NUMBER(5);
	PER NUMBER(7,2);
	GRADE VARCHAR2(30);
BEGIN
	ROLL_NO:=&ROLL_NO;
	NAME:='&NAME';
	SUB1:=&SUB1;
	SUB2 :=&SUB2 ;
	SUB3 :=&SUB3 ;
	SUB4:=&SUB4;
	SUB5:=&SUB5;
	TOTAL:=SUB1+SUB2+SUB3+SUB4+SUB5;
	PER:=TOTAL/5;

	IF (PER>=75) THEN
		GRADE:='O';
	ELSIF (PER<=75)AND(PER>=70) THEN
		GRADE:='A';
```

```
        ELSIF (PER<=60)AND(PER>=70) THEN
                GRADE:='B';
        ELSIF (PER<=50)AND(PER>=60) THEN
                GRADE:='C';
        ELSE
                GRADE:='PASS CLASS';
        END IF;

        INSERT INTO
RESULT(ROLLNO,NAME,SUB1,SUB2,SUB3,SUB4,SUB5,TOTAL,PER,GRADE)VALUES(RO
LL_NO,NAME,SUB1,SUB2,SUB3,SUB4,SUB5,TOTAL,PER,GRADE);

        DBMS_OUTPUT.PUT_LINE('    ');
        DBMS_OUTPUT.PUT_LINE('Roll no:-'||ROLL_NO);
        DBMS_OUTPUT.PUT_LINE('Name:-'||NAME);
        DBMS_OUTPUT.PUT_LINE('    ');
        DBMS_OUTPUT.PUT_LINE('---------------------Result--------------------------');
        DBMS_OUTPUT.PUT_LINE('Marks of sub1:-'||SUB1);
        DBMS_OUTPUT.PUT_LINE('Marks of sub2:-'||SUB2);
        DBMS_OUTPUT.PUT_LINE('Marks of sub3:-'||SUB3);
        DBMS_OUTPUT.PUT_LINE('Marks of sub4:-'||SUB4);
        DBMS_OUTPUT.PUT_LINE('Marks of sub5:-'||SUB5);

        DBMS_OUTPUT.PUT_LINE('    ');
        DBMS_OUTPUT.PUT_LINE('---------------------------------------------');
        DBMS_OUTPUT.PUT_LINE('Your total marks : '||TOTAL);
        DBMS_OUTPUT.PUT_LINE('Percentage :'||PER||'%');
        DBMS_OUTPUT.PUT_LINE('Grade:'||GRADE);
END;
/

SET SERVEROUTPUT OFF
SET VERIFY ON
SET FEEDBACK ON
```

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p3.sql
Enter value for roll_no: 17
Enter value for name: kunjal
Enter value for sub1: 80
Enter value for sub2: 90
Enter value for sub3: 88
Enter value for sub4: 96
Enter value for sub5: 95

Roll no:-17
Name:-kunjal

---------------------Result--------------------------
Marks of sub1:-80
Marks of sub2:-90
Marks of sub3:-88
Marks of sub4:-96
Marks of sub5:-95

-----------------------------------------------
Your total marks : 449
Percentage :89.8%
Grade:O
```

4. 1. Write a program that prints value 1 to 100 numbers using FOR LOOP.
2. Write a program that prints value 1 to 100 number using LOOP Command.
3. Write a program that prints value 1 to 100 number using WHILE LOOP Command

---

```
/*
1. Write a program that prints value 1 to 100 numbers using FOR LOOP.
2. Write a program that prints value 1 to 100 number using LOOP
Command.
3. Write a program that prints value 1 to 100 number using WHILE LOOP
Command
*/

SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF

DECLARE
  i NUMBER := 1;
BEGIN

  -- FOR LOOP
  DBMS_OUTPUT.PUT_LINE('--- FOR LOOP ---');
  FOR i IN 1..100 LOOP
    DBMS_OUTPUT.PUT(i || ' ');
  END LOOP;
  DBMS_OUTPUT.PUT_LINE('                          ');


  -- SIMPLE LOOP
  DBMS_OUTPUT.PUT_LINE('--- SIMPLE LOOP ---');
  i := 1;
  LOOP
    DBMS_OUTPUT.PUT(i || ' ');
    i := i + 1;
    EXIT WHEN i > 100;
  END LOOP;
  DBMS_OUTPUT.PUT_LINE('');
```

```
 DBMS_OUTPUT.PUT_LINE('--- WHILE LOOP ---');
  i := 1;
WHILE i <= 100 LOOP
        DBMS_OUTPUT.PUT(i || ' ');
        i := i + 1;
    END LOOP;
      DBMS_OUTPUT.PUT_LINE('            ');
END;
/
SET SERVEROUTPUT OFF
SET VERIFY ON
SET FEEDBACK ON
```

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p4.sql
--- FOR LOOP ---
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57
58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84
85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
--- SIMPLE LOOP ---
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57
58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84
85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
--- WHILE LOOP ---
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57
58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84
85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
```

5. Write a PL/SQL block which displays gross salary of employees as per user input EID. (Consider EMP table with EID, EName, Deptno,Deptname Gender, Age, BasicSal) with appropriate data types.) Gross_Salary: BASICSAL + (DA + HRA + Medical) – PF. Rules: HRA = 15% of basic, DA = 50% of basic, Medical = Rs. 500, PF = 10% of basic.

```
/*
Write a PL/SQL block which displays gross salary of employees as per user input
EID.
Consider EMP table with EID, EName, Deptno, Deptname, Gender, Age, BasicSal
with appropriate data types.
Gross_Salary:= BASICSAL + (DA + HRA + Medical) – PF.
Rules: HRA = 15% of basic, DA = 50% of basic, Medical = Rs. 500, PF = 10% of basic.
Here for practical we have used Scott users inbuilt EMP table.
*/

SET SERVEROUTPUT ON
SET FEEDBACK OFF
SET VERIFY OFF

DECLARE
   V_EID       EMP.EID%TYPE;
   V_NAME      EMP.ENAME%TYPE;
   BASIC_SAL   EMP.BASICSAL%TYPE;
   DA          NUMBER;
   HRA         NUMBER;
   MEDICAL     NUMBER := 500;
   PF          NUMBER;
   GROSS_SAL   NUMBER;

BEGIN
   V_EID := &EID;

   SELECT ENAME, BASICSAL
   INTO V_NAME, BASIC_SAL
   FROM EMP
   WHERE EID = V_EID;

   DA := BASIC_SAL * 0.50;
   HRA := BASIC_SAL * 0.15;
```

```
  PF := BASIC_SAL * 0.10;
  GROSS_SAL := BASIC_SAL + DA + HRA + MEDICAL - PF;

  DBMS_OUTPUT.PUT_LINE('--------------- SALARY SLIP ---------------');
  DBMS_OUTPUT.PUT_LINE('Employee ID     : ' || V_EID);
  DBMS_OUTPUT.PUT_LINE('Employee Name   : ' || V_NAME);
  DBMS_OUTPUT.PUT_LINE('Basic Salary    : Rs. ' || BASIC_SAL);
  DBMS_OUTPUT.PUT_LINE('DA Amount       : Rs. ' || DA);
  DBMS_OUTPUT.PUT_LINE('HRA Amount      : Rs. ' || HRA);
  DBMS_OUTPUT.PUT_LINE('Medical Amount  : Rs. ' || MEDICAL);
  DBMS_OUTPUT.PUT_LINE('Provident Fund  : Rs. ' || PF);
  DBMS_OUTPUT.PUT_LINE('--------------------------------------------');
  DBMS_OUTPUT.PUT_LINE('Gross Salary    : Rs. ' || GROSS_SAL);
  DBMS_OUTPUT.PUT_LINE('--------------------------------------------');
EXCEPTION
  WHEN NO_DATA_FOUND THEN
    DBMS_OUTPUT.PUT_LINE(
      'Employee record for given EID - ' || V_EID || ' does not exist'
    );
END;
/

SET SERVEROUTPUT OFF
SET FEEDBACK ON
SET VERIFY ON
```

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p5.sql
Enter value for eid: 105
--------------- SALARY SLIP ---------------
Employee ID     : 105
Employee Name   : PRIYA
Basic Salary    : Rs. 32000
DA Amount       : Rs. 16000
HRA Amount      : Rs. 4800
Medical Amount  : Rs. 500
Provident Fund  : Rs. 3200
--------------------------------------------
Gross Salary    : Rs. 50100
--------------------------------------------
```

6. Write a PL/SQL block which accepts measurement in feet and displays it in cm, inch and meter.

```
/*
Write a PL/SQL block which accepts measurement in feet and displays it
in cm, inch and meter.

Formulas:-
        1)Inches :-feet * 12
        2)Centimeters :- feet * 30.48
        3)Meters:-feet*0.3045
*/

SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF

DECLARE
   FEET   NUMBER;
   CM     NUMBER;
   INCH   NUMBER;
   METER  NUMBER;
BEGIN
   FEET := &FEET;

   CM := FEET * 30.48;
   INCH := FEET * 12;
   METER := FEET * 0.3048;

   DBMS_OUTPUT.PUT_LINE('Feet      : ' || FEET);
   DBMS_OUTPUT.PUT_LINE('Centimeter : ' || CM);
   DBMS_OUTPUT.PUT_LINE('Inch      : ' || INCH);
   DBMS_OUTPUT.PUT_LINE('Meter     : ' || METER);
END;
/

SET SERVEROUTPUT OFF
SET VERIFY ON
SET FEEDBACK ON
```

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p5.sql
Enter value for eid: 105
--------------- SALARY SLIP ---------------
Employee ID      : 105
Employee Name    : PRIYA
Basic Salary     : Rs. 32000
DA Amount        : Rs. 16000
HRA Amount       : Rs. 4800
Medical Amount   : Rs. 500
Provident Fund   : Rs. 3200
--------------------------------------------
Gross Salary     : Rs. 50100
--------------------------------------------
```

---

7. Write a PL/SQL block which converts temperature from Celsius to Fahrenheit.

```
/*
Write a PL/SQL block which converts temperature from Celsius to
Fahrenheit.
Formulas:-
        -Celsius to Fahrenheit :-(Celsius *9/2)+32
        -Fahrenheit to Celsius :-(Fahrenheit-32)/5/9
*/

SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF
ACCEPT CH PROMPT 'Enter your choice :- 1. Celsius to Fahrenheit, 2. Fahrenheit to
Celsius: '
DECLARE
        CELSIUS NUMBER(10,2);
        FAHRENHEIT NUMBER(10,2);
        CHOICE NUMBER(10);
BEGIN
   CHOICE := &CH;
   IF CHOICE = 1 THEN
      CELSIUS := &CELSIUS;
      FAHRENHEIT := (CELSIUS * 9 / 5) + 32;

     DBMS_OUTPUT.PUT_LINE('Value of FAHRENHEIT :- ' || FAHRENHEIT);

   ELSIF CHOICE = 2 THEN
         FAHRENHEIT := &FAHRENHEIT;
         CELSIUS := (FAHRENHEIT - 32) * 5 / 9;
      DBMS_OUTPUT.PUT_LINE('Value of CELSIUS :- ' || CELSIUS ||'C');
   ELSE
      DBMS_OUTPUT.PUT_LINE('INVALID CHOICE... Enter 1 or 2');
   END IF;
END;
/
```

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p7.sql
Enter your choice :- 1. Celsius to Fahrenheit, 2. Fahrenheit to Celsius: 1
Enter value for celsius: 12
Enter value for fahrenheit: 30
Value of FAHRENHEIT :- 53.6
```

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p7.sql
Enter your choice :- 1. Celsius to Fahrenheit, 2. Fahrenheit to Celsius: 2
Enter value for celsius: -45
Enter value for fahrenheit: 45
Value of CELSIUS :- 7.22C
SQL>
```

---

8. Write a PL/SQL block to delete the record of employee for given EID.

```
/*
Write a PL/SQL block to remove an employee record
for the given Employee ID.
*/

SET SERVEROUTPUT ON
SET FEEDBACK OFF
SET VERIFY OFF
DECLARE
   EMP_ID EMP.EID%TYPE;
   CNT NUMBER;
BEGIN
   EMP_ID := &EID;

   SELECT COUNT(*) INTO CNT FROM EMP WHERE EID = EMP_ID;

   IF CNT > 0 THEN
      DELETE FROM EMP WHERE EID = EMP_ID;

      COMMIT;
      DBMS_OUTPUT.PUT_LINE( 'Employee ID ' || EMP_ID || ' deleted successfully.'
);
   ELSE
      DBMS_OUTPUT.PUT_LINE('Employee ID ' || EMP_ID || ' does not exist.' );
   END IF;

END;
/
SET SERVEROUTPUT OFF
SET FEEDBACK ON
SET VERIFY ON
```

Output:

9. Write a PL/SQL block that calculates the simple interest based on given principal amount, rate of interest and number of years.

```
/* Write a PL/SQL block to calculate Simple Interestand Total Amount. */

SET SERVEROUTPUT ON
SET FEEDBACK OFF
SET VERIFY OFF
DECLARE
   P NUMBER;
   R NUMBER;
   T NUMBER;
   SI NUMBER;
   AMOUNT NUMBER;
BEGIN
   P := &P;
   R := &R;
   T := &T;
   SI := (P * R * T) / 100;
   AMOUNT := P + SI;
  DBMS_OUTPUT.PUT_LINE('=======================');
   DBMS_OUTPUT.PUT_LINE('Principal Amount : ' || P);
   DBMS_OUTPUT.PUT_LINE('Rate of Interest : ' || R || '%');
   DBMS_OUTPUT.PUT_LINE('Number of Years  : ' || T);
   DBMS_OUTPUT.PUT_LINE('Simple Interest  : ' || SI);
   DBMS_OUTPUT.PUT_LINE('Total Amount     : ' || AMOUNT);
END;
/
SET SERVEROUTPUT OFF
SET FEEDBACK ON
SET VERIFY ON
```

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p9.sql
Enter value for p: 12000
Enter value for r: 10
Enter value for t: 5
========================
Principal Amount : 12000
Rate of Interest : 10%
Number of Years  : 5
Simple Interest  : 6000
Total Amount     : 18000
```

10. Write a PL/SQL block to calculate the square and cube of the given number.

```
/*
Write a PL/SQL block to calculate the square and cube
of the given number.
*/

SET SERVEROUTPUT ON
SET FEEDBACK OFF
SET VERIFY OFF

DECLARE
   N NUMBER;
   SQUARE NUMBER;
   CUBE NUMBER;
BEGIN
   N := &N;
   SQUARE := N * N;
   CUBE := N * N * N;

   DBMS_OUTPUT.PUT_LINE('Given Number : ' || N);
   DBMS_OUTPUT.PUT_LINE('Square       : ' || SQUARE);
   DBMS_OUTPUT.PUT_LINE('Cube         : ' || CUBE);

END;
/

SET SERVEROUTPUT OFF
SET FEEDBACK ON
SET VERIFY ON
```

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p10.sql
Enter value for n: 20
Given Number : 20
Square       : 400
Cube         : 8000
SQL>
```